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School of Mathematical Sciences
Department of Computer Science

MSc Computer Science : Batch 2024-26, Semester I, Midterm
DA104: Probability and Stochastic processes
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*S*tudent Name (in block letters):	Date: 20 September 2024
*S*tudent Roll No:	*M*ax Marks: 60
*S*ignature:	Time: 2hrs

Answer the following questions

1. State the three axioms of probability. Using the axioms prove that $P(\phi) = 0$. [3+2]

2. Consider a well shuffled deck of 52 cards and suppose we draw three cards at random. What is the probability that at least one is an ace? [5]

3. Drawers I and II contain black and red pencils as follows:
   Drawer I: b_1 black pencils + r_1 red pencils.
   Drawer II: b_2 black pencils + r_2 red pencils.
   If a drawer is chosen at random and then a pencil is chosen at random from the drawer, then:
   a) Compute the probability that the pencil chosen is red. [2]
   b) Given that the chosen pencil is black, what is the probability that drawer II was chosen? [3]

4. In a certain book of N pages, no page contain more than four errors, n_1 of them contain one error, n_2 contain two errors, n_3 contain three errors and n_4 contain four errors. Two copies of the book are taken and a page is opened at random from each book. What is the probability that the total number of errors on both the pages does not exceed five? [5]

5. An urn contains n balls numbered 1 through n. You draw m balls in sequence, each time replacing the ball selected previously. Suppose X, a random variable, denotes the minimum number on the ball among the m balls chosen. Compute the p.m.f of X. [5]

6. A casino patron will continue to make $5 bet on red in roulette until he has won 4 of these bets. Probability of winning a bet is p.
   a) What is the probability that he places a total of 9 bets? [3]
   b) What is the expected winnings if the expected number of games played is S. [2]

7. Suppose X is a continuous random variable with density function $f(x)$ defined:

$$f(x) = \begin{cases} cx^4 & 0 < x < 2 \\ \frac{x+1}{4} & 1 \le x < 3 \\ 0 & \text{otherwise} \end{cases}$$

   Compute $E(X)$ and $Var(X)$. [3+3]

8. The random variable X has probability density function

$$f(x) = \begin{cases} ax + bx^2 & 0 < x < 1 \\ \frac{x+1}{4} & 1 \le x < 3 \\ 0 & \text{otherwise} \end{cases}$$

If $E[X] = 0.6$ find $P(X < 0.5)$ [4]

9. If X is a non-negative random variable then prove that $E(\log X) \leq \log(E[X])$. [4]

10. Prove that $[E(XY)]^2 \leq E(X^2)E(Y^2)$. [6]

11. Two r.v.'s X and Y have joint p.d.f

$$f_{x,y}(x, y) = \frac{6}{7}(x^2 + \frac{xy}{2}), 0 < x \leq 1, 0 < y \leq 2.$$

a) Show that $f_{x,y}$ is indeed a p.d.f. [5]

b) Calculate $P(X > Y)$. [5]